Requirement Analysis Report

Project Title

E-Commerce Order Management Database System

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1. Introduction

Overview

The E-Commerce Order Management Database System is designed to manage the daily operations of an online shopping platform. The system stores and manages customer information, product details, supplier records, inventory, customer orders, payments, shipment tracking, and product reviews. A centralized database improves data accuracy, reduces redundancy, and supports efficient business operations.

Objective

The main objective of this project is to analyze the business requirements of an e-commerce company and prepare a structured requirement specification before designing the database.

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2. Business Scenario Analysis

A rapidly growing e-commerce company requires a database system to handle its increasing number of customers and orders. The system should allow customers to register, browse products, place orders, make payments, track shipments, and provide product reviews.

The company also needs to manage product categories, suppliers, inventory, payments, and shipment information. A centralized database ensures that all business information is stored securely and efficiently while reducing duplicate data.

The proposed database system will improve business operations by:

- Managing customer information efficiently.
- Maintaining accurate product and inventory records.
- Processing customer orders quickly.
- Recording payment transactions securely.
- Tracking product deliveries.
- Collecting customer feedback through reviews.
- Generating business reports for management.

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3. Purpose of the System

The purpose of the E-Commerce Order Management Database System is to provide an integrated platform for managing all business operations of an online shopping company. The system helps automate daily activities, maintain accurate records, and improve customer satisfaction.

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4. Core Entities and Their Purpose

1. Customer

The Customer entity stores the personal details of registered customers. It includes Customer ID, Name, Email, Mobile Number, Address, and Registration Date. This information is used during order processing and customer communication.

2. Category

The Category entity stores different product categories such as Electronics, Fashion, Books, and Home Appliances. It helps organize products efficiently.

3. Product

The Product entity stores detailed information about products available for sale. It includes Product ID, Product Name, Price, Stock Quantity, and Category ID.

4. Supplier

The Supplier entity maintains supplier information, including Supplier ID, Supplier Name, and Contact Information. Suppliers provide products to the company.

5. Order

The Order entity records every purchase made by customers. It stores Order ID, Customer ID, Order Date, Total Amount, and Order Status.

6. Order Details

The Order Details entity stores information about each product included in an order. It contains Order Detail ID, Order ID, Product ID, Quantity, and Unit Price.

7. Payment

The Payment entity stores payment transaction details such as Payment ID, Order ID, Payment Method, Payment Date, and Payment Status.

8. Shipment

The Shipment entity records delivery information including Shipment ID, Order ID, Delivery Address, Shipment Date, and Delivery Status.

9. Review

The Review entity stores customer feedback for purchased products. It contains Review ID, Customer ID, Product ID, Rating, and Comments.

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5. Major Business Processes

The system performs the following business processes:

1. Customer Registration
2. Customer Login
3. Product Browsing
4. Product Category Management
5. Inventory Management
6. Order Placement
7. Order Processing
8. Payment Processing
9. Shipment Tracking
10. Customer Review Management
11. Report Generation

These processes help the organization manage daily operations efficiently.

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6. Conclusion

The Requirement Analysis identifies the major business needs of the proposed E-Commerce Order Management Database System. It provides a clear understanding of the system requirements and serves as the foundation for designing the database in the upcoming phases.
